DOCUMENT 2 OF 3 · HIGH LEVEL DESIGN

System Architecture

High Level Design. C4 levels 1 and 2, runtime flows, technology decisions and compliance architecture.

Version 0.1 · Draft for engineering review

Implements the Product Requirements Document, version 0.1

14 September 2026

SECTION 1

Purpose and principles

This document describes the system at C4 levels 1 and 2 — the context it sits in and the containers it deploys as. Component-level structure, schemas and API contracts are in the Low Level Design.

Architectural principles

Principle	Consequence
Clinical logic is configuration	Thresholds, bands, weights and detectors live in versioned config, not code. A surgeon changes behaviour without a deployment, and every decision records the ruleset version that produced it.
The evaluator is a pure function	Reading, baseline, history and config in; tier and reasons out. No input or output of any kind. This is the one component that must be auditable and exhaustively testable in isolation.
Synchronous evaluation	The patient's device must learn immediately whether to display emergency guidance. Evaluation therefore happens in the request path, not in a queue.
Absence is a signal	Missing data is never treated as normal. A patient who stops submitting has a distinct status, and detecting that requires a scheduled process rather than a request handler.
Build for the scale that exists	A few hundred patients submitting once daily is a negligible write volume. No broker, no cache, no service decomposition. Operational surface area is the cost that matters at this stage.

SECTION 2

C1 — System context

Who uses the system and what it depends on, with no internal detail.

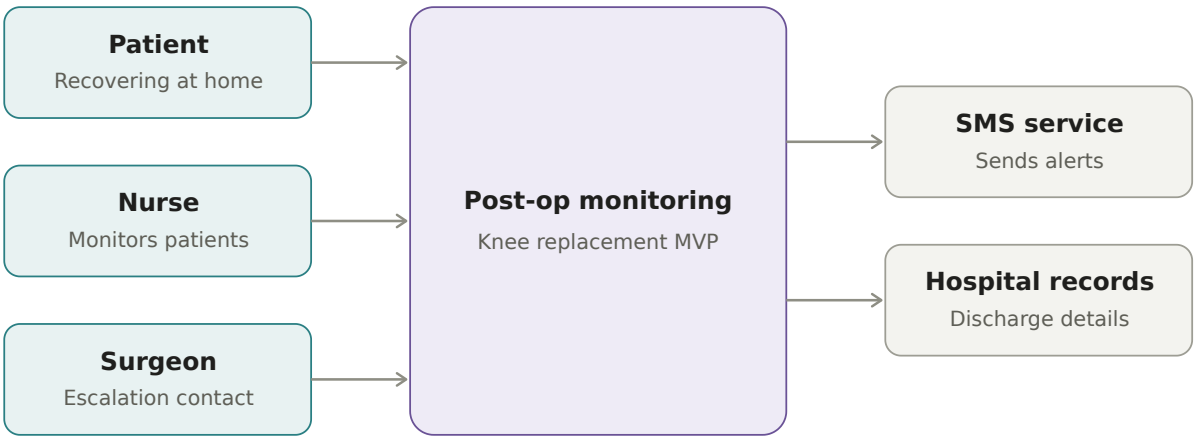


Figure 1. C4 level 1 — system context.

Element	Type	Responsibility
Patient	Person	Completes the daily check-in, captures the wound photograph, answers the nurse's call. Typically 55 to 75, mid-range Android handset, possibly assisted by a family member.
Nurse	Person	Works the risk-sorted queue, acknowledges alerts, calls patients, records outcomes. Primary daily user.
Surgeon	Person	Approves the ruleset and per-patient thresholds at discharge. Receives escalations on unacknowledged critical alerts. Decides on same-day review.
Monitoring system	System (in scope)	Collects check-ins, evaluates against limits, surfaces concerning cases by urgency, and guarantees no critical alert goes unacknowledged.
SMS service	External	Delivers critical alerts to on-call staff and reminders to patients. Chosen because it does not depend on the recipient having the app open or notifications enabled.
Hospital records	External	Source of discharge baseline and demographics. **Optional in the MVP** — replaced by manual enrolment.

> **ON THE HOSPITAL RECORDS DEPENDENCY**
>
> Drawn as a real dependency because it is the correct target state, but deliberately deferred. Integration is a months-long procurement conversation with hospital IT. The MVP substitutes a nurse typing in name, surgery date and discharge readings. Nothing in the architecture assumes the integration exists.

SECTION 3

C2 — Container view

The deployable units and how data moves between them. Six containers is what the MVP requires and no more.

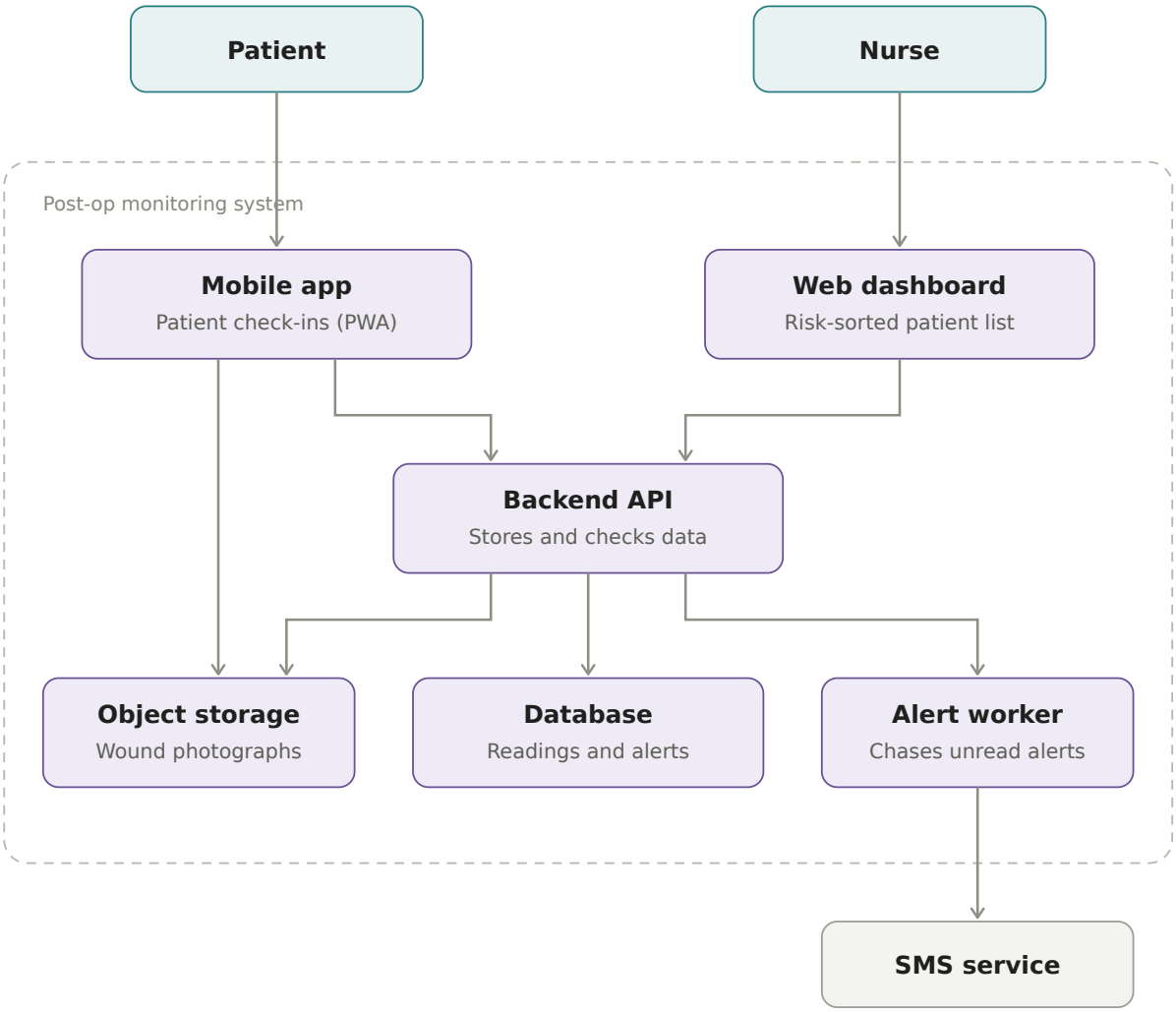


Figure 2. C4 level 2 — containers. The mobile app writes photographs directly to object storage using a signed URL, bypassing the API.

Container	Technology	Responsibility and rationale
Mobile app	Progressive web app; React Native only if native camera control proves necessary	Reminder, check-in form, photo capture, offline queue, emergency guidance banner. A PWA sidesteps app-store friction and install reluctance in an older cohort.
Web dashboard	React single-page application	Risk-sorted list, patient detail and trends, acknowledge, record outcome. Desktop-first; nurses use it at a station.
Backend API	FastAPI (Python) or Express (Node)	Authentication, check-in ingest, rule evaluation, dashboard queries, alert creation. Python is the safer choice if trend or image work is added later.
Database	PostgreSQL	Patients, readings, evaluations, alerts, outcomes, rulesets. Relational fits cleanly; data is small, structured, query patterns known.
Object storage	S3-compatible	Wound photographs. Kept out of the database for size and for signed-URL access control.
Alert worker	Scheduled job (cron, APScheduler or Celery beat)	Escalation timer for unacknowledged critical alerts; daily sweep for missed check-ins. The only genuinely time-driven behaviour, and it cannot live in a request handler because nothing triggers it.

SECTION 4

Runtime flows

4.1 Daily check-in and evaluation

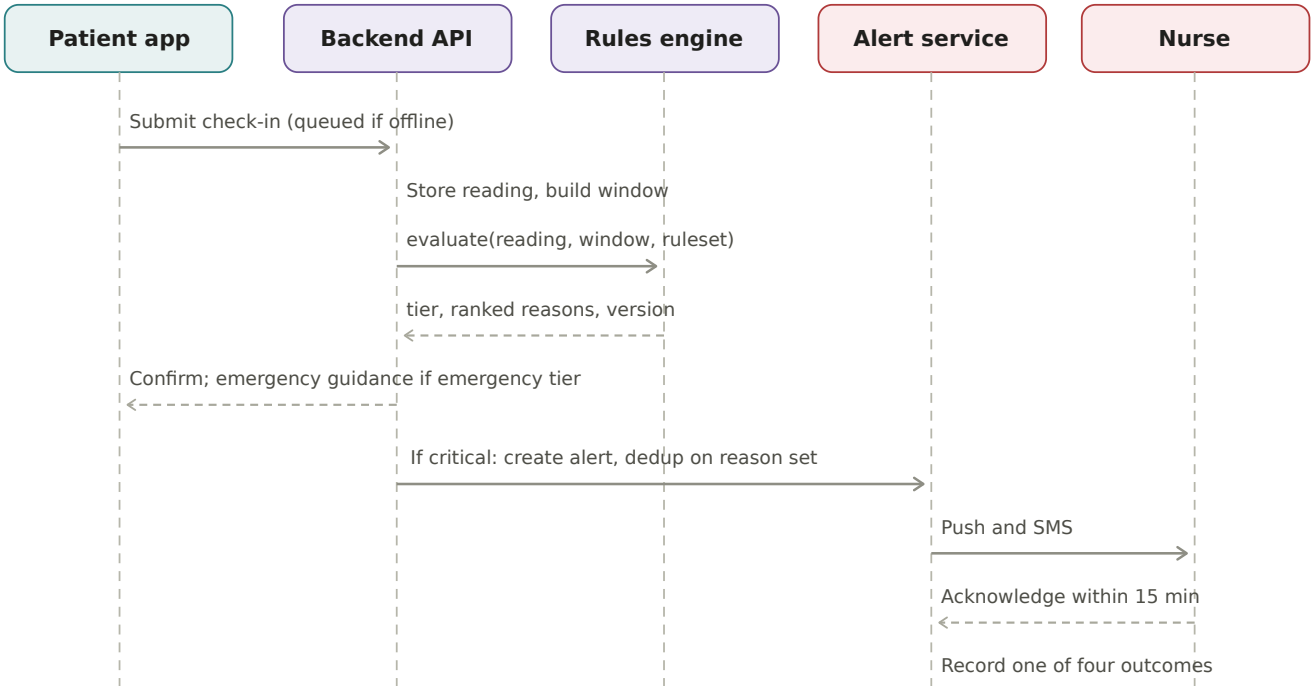


Figure 3. The synchronous evaluation path. The patient's confirmation does not wait on alert dispatch.

Evaluation is synchronous and deliberately so: the response tells the device whether to display emergency guidance. Alert creation and notification dispatch happen after the response is committed, so a slow SMS provider cannot delay the patient's confirmation.

4.2 Escalation sweep

The alert worker runs on a short interval, typically every minute. It selects critical alerts whose status is still open and whose notification timestamp is older than the configured acknowledgement window, then escalates each to the backup nurse or on-call physician and records the escalation. Escalation is idempotent: an already-escalated alert is skipped.

4.3 Missed check-in sweep

A daily job identifies patients whose expected check-in for the current cadence has not arrived. For each, it invokes the same evaluator with a synthetic reading in which every field is unknown, so that absence detectors fire through the normal path. This is the reason evaluation must be callable from two places, and why a parallel implementation inside the worker would be a defect rather than an optimisation.

4.4 Photograph upload

The app requests a short-lived signed upload URL from the API, writes the image directly to object storage, and submits only the resulting key with the check-in. Images never transit the API. The dashboard reads them through short-lived signed download URLs.

SECTION 5

Cross-cutting architecture

5.1 Data architecture

Six principal entities: patients with their discharge baseline, readings, evaluations, alerts with outcomes, clinicians, and versioned rulesets. Readings are append-only; the evaluator uses the latest submission for a given post-operative day but earlier submissions are retained for audit. Evaluations store the ruleset version and the ranked contributing signals, which is what makes a past decision reconstructable. Full schema is in the Low Level Design.

5.2 Alerting architecture

Two-stage by design. Hard overrides are checked first and bypass scoring entirely; only then does a weighted sum run across the remaining signals. A single weighted total is insufficient on its own for three reasons: a critical individual finding can be diluted below threshold by good scores elsewhere, a numeric total gives a nurse nothing actionable, and false positives become untraceable to a specific weight. The design mirrors established early-warning score practice, where any single parameter at its maximum escalates irrespective of the aggregate.

5.3 Authentication and authorisation

Actor	Mechanism
Patient	One-time passcode by SMS against the enrolled number, exchanged for a long-lived scoped token. No password at any point, because password management is an unreasonable burden on this cohort and a support cost.
Clinician	Email and password with a second factor, short-lived session, role-based access distinguishing nurse from surgeon from administrator.
Alert worker	Service credential, no interactive session, restricted to the alert and reading tables.

5.4 Compliance architecture

- **Consent.** Captured at enrolment against a versioned consent text, stored with timestamp and purpose. Withdrawal marks the patient inactive and halts all processing and notification.
- **Purpose limitation.** Data is collected for post-operative safety monitoring only. No secondary use without separate consent.
- **Retention.** Readings and evaluations retained per clinical record requirements; photographs subject to a shorter, separately configured retention.
- **Breach process.** Documented and rehearsed, capable of notifying the Data Protection Board and affected patients within seventy-two hours.
- **Audit.** Every clinical decision, acknowledgement and outcome recorded immutably with actor and timestamp.

5.5 Deployment

Three environments: local development with seeded fixture patients, a staging environment carrying no real patient data, and production hosted in an Indian region. Containerised deployment with managed PostgreSQL and managed object storage. Continuous integration runs the full fixture-patient suite on every change to the ruleset or the evaluator; a change to either that alters any fixture's first-escalation day fails the build.

5.6 Failure modes

Failure	Behaviour
Device offline at check-in time	Submission queued locally with original capture time, retried on reconnection. Up to seven days retained.
SMS provider unavailable	Notification retried with backoff; failure after retries escalates as if unacknowledged. Push notification remains an independent channel.
Alert worker stops	Escalation ceases silently, which is the most dangerous failure in the system. Requires a dedicated liveness check and alarm independent of the worker itself.
Object storage unavailable	Check-in still accepted without the photograph. The submission is not blocked on an image.
Evaluation error	Reading is stored and flagged for manual review rather than silently discarded. A patient must never disappear because of a code fault.

SECTION 6

Decisions and evolution

Architecture decision record

ID	Decision	Rationale and consequence
AD-1	Progressive web app rather than native	Avoids app-store friction and install reluctance in a 55 to 75 cohort. Accepts less reliable push notification, mitigated by SMS fallback. Revisit if photo quality proves inadequate.
AD-2	Synchronous evaluation in the request path	Emergency guidance must reach the patient immediately. Constrains the evaluator to be fast and free of input or output.
AD-3	Rules as versioned configuration	Enables surgeon-driven change without deployment and makes past decisions reconstructable. Costs a configuration schema and validation layer.
AD-4	Separate scheduled worker	Escalation and absence detection are time-driven and have no request to attach to. Adds a second deployable and a liveness monitoring obligation.
AD-5	No message broker or cache	Write volume is negligible. Adding either buys operational burden and nothing else at this scale.
AD-6	Manual enrolment instead of EHR integration	Removes a procurement dependency from the critical path. Costs nurse time per patient and risks transcription error in the baseline.
AD-7	Object storage separate from the database	Photographs are large and need signed-URL access control. Adds a container and a lifecycle policy.
AD-8	Patient-facing risk content limited to emergency guidance	Prevents anxiety-driven contact. Means the app cannot explain why a nurse is calling.

Evolution path beyond the MVP

1. **Second procedure type.** Requires only a new ruleset and form schedule, provided nothing procedure-specific has leaked into code. This is the first real test of AD-3.
2. **Multi-tenancy.** Clinic scoping exists in the schema from the start but is not exercised. Adds role and data isolation work.
3. **EHR integration.** Replaces manual enrolment. Introduces FHIR parsing and a hospital IT dependency; worth parsing plausible FHIR bundles in testing before the conversation begins.
4. **Composite early-warning scoring.** A natural upgrade once the rule-based version has demonstrable behaviour to compare against.
5. **Wound image analysis.** Only after a labelled photograph corpus exists, which the MVP is quietly accumulating.